James Guillochon

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Publications: **126** (**27** first author w/ **2,579** citations, h-index 14)

Education

Harvard — Postdoctoral Scholar, Astronomy — Cambridge, MA **June 2018**

Insight Data Science — Fellow, Data Science — Boston, MA **June 2018**

UC Santa Cruz — PhD, Astronomy — Santa Cruz, CA **June 2013**

UC Irvine — BS, Physics — Irvine, CA **June 2005**

Work Experience

Esri — Principal Data Scientist San Diego, CA (Home Office) **Jan 2021 - Present (5 years)**

- **Leadership & AI Strategy:** Co-led the AI & Analytics Tech Center, shaped Professional Services AI strategy and customer-readiness materials, mentored staff, and supported bids, workshops, and customer scoping.
- **Generative AI & LLM Tooling:** Contributed to generative AI workflows by implementing tools that integrate large language models for image interrogation, free-form text object detection, and automated object replacement.
- **Deep Learning Imagery Analysis:** Built deep learning pipelines for satellite and aerial imagery covering zero-shot detection, land classification, algal-bloom prediction, feature extraction, and time-series monitoring.
- **3D Scene & LIDAR Optimization:** Optimized 3D LIDAR processing pipelines and applied PointCNN models alongside advanced AI algorithms and image embeddings to identify infrastructural assets within 3D scenes.
- **Agentic AI Harnesses:** Developed multiplatform agentic harnesses with MCP servers and CLIs for model serving, image AI proxying, and tool orchestration across local and cloud developer workflows.
- **Oriented Imagery Asset Extraction:** Delivered oriented-imagery and embedding-based tools that extract infrastructural assets from street-level and oblique imagery to support transportation and utility workflows.
- **Scanned Document Asset Extraction:** Built reusable document AI tools that extract assets and mapped extents from scanned records, combining OCR, language models, and geospatial enrichment for customer delivery.
- **Automated Anomaly Detection:** Designed and deployed automated solutions to actively monitor open-source map catalogs and identify suspicious changesets.
- **NLP & Document Parsing:** Authored parsing tools to analyze scanned legal documents and automatically extract and map the geographic regions described within the text.
- **PS AI Enablement Tooling:** Built reusable catalogs and assistants for staff skills discovery, engagement battle cards, code-sample reuse, and customer-needs drafting to accelerate proposal and delivery work.

Berkshire Grey — Principal Data Scientist Lexington, MA **Oct 2018 - Dec 2020 (2 years)**

- **Robotics Software Stack:** Led optimization efforts of Python/ROS software stack used for pick and place in a multi-component robotics system, including robotic arms, high-performance conveyors, and automated shuttles.
- **Motion Planning & Parameter Prediction:** Created and integrated parameter prediction to select optimal robotic motions for unseen products on demand.
- **Bayesian Sensing & Pick Verification:** Developed online Bayesian analysis of data from sensors integrated into the picking system to dynamically determine how many items the robotic arm lifted.
- **Operations Analytics:** Curated a dashboard of key performance indicators (KPIs) for the deployed robotic system via the ELK Stack.

Harvard-Smithsonian — NASA Einstein/ITC Fellow Cambridge, MA **Sept 2013 - Aug 2018 (5 years)**

- **Research Leadership:** Technical lead of an eight-person research team of undergraduate/graduate students and postdocs resulting in dozens of completed scientific projects with 10k+ citations.
- **Science Communication:** Appeared on the NOVA television program in 2018, describing data analysis by my group to measure a black hole's mass using a recently destroyed star.
- **MOSFiT & Bayesian Modeling:** Developed MOSFiT, a Bayesian engine to infer the most likely analytical models for time-series data.
- **Open Astronomy Catalogs:** Led creation of the Open Astronomy Catalogs, a platform for sharing astronomical data, with a user-facing frontend and an API backend to serve data quickly to users.